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Linguistics in a battlefield.
A short note on syntax and the "Newtonian style of research".[1] (v1)

Abstract. Linguistics has always been a battlefield as the vision of language has always been the expression of the vision of human beings in any culture. Nowadays, neurobiological models and computational theories compete with linguistics towards the understanding of the origin, the function and the structure of human languages. These notes highlight two crucial points: first, these two competitors target two distinct and incompatible goals; second, there are aspects of human languages, like the systematic differences expressed within the boundaries of Babel and crucial notions such as "predication" which are unformulable outside linguistics theory.

1. An inherent battlefield with three actors.

Linguistics has always been a battlefield as the vision of language has always been the expression of the vision of human beings in any culture. It is hardly surprising then that nowadays linguistics is still riven by strong tensions. What is surprising is that the major reasons for these tensions come from the strains generated by two, so to speak, external forces, namely the domain of neurobiological studies on the one hand and computational theories on the other. These two domains are obviously characterized by different goals and different methods although they share the same object, i.e. human language.

The possibility of relating human language to brain activity was obviously established in a scientific sense at least with the pioneering work of Paul Broca in the second half of the XIX century but it should be highlighted that this connection referred to the general capacity to speak and understand language, hence to language as a human faculty, rather than any specific linguistic structure or any specific language (Cappa et al. 2000). Moreover, the philosophical mainstream of analytic works, such as those paradigmatically expressed by Ludwig Wittgenstein as well as by Willard Van Orman Quine, constituted a dramatic obstacle to any further biological investigation of language structure. The worried caveat addressed by Eric Lenneberg's author–itative words in the introduction to his famous volume on "The Biological Foundations of Language" witnesses this attitude and at the same time it highlights the profound change of paradigm in a very clear way: "A biological investigation into language must

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seem paradoxical as it is so widely assumed that languages consist of arbitrary, cultural conventions. Wittgenstein and his followers speak of the word game, thus likening languages to the arbitrary set of rules encountered in parlor games and sports. It is acceptable usage to speak of the psychology of bridge or poker, but a treatise on the biological foundation of contract bridge would not seem to be an interesting topic. The rules of natural languages do bear some superficial resemblance to the rules of a game, but I hope to make it obvious in the following chapters that there are major and fundamental differences between rules of languages and rules of games. The former are biologically determined; the latter are arbitrary.'' (Lenneberg 1967: 2).

Lenneberg's successful research especially focused on syntax, the recognized core of all and only human languages, marks a dramatic paradigm shift and one with far reaching consequences: some thirty years later, the first experimental, i.e. quantitative, analyses correlating syntactic computation with selective brain networks and their corresponding metabolism could be carried out (Moro et al. 2001, Tettamanti et al. 2001, Musso et al. 2003, Moro 2008/2015, Moro 2016, Friederici 2017, Artoni et al. 2020). Crucially, Lenneberg's results could not be interpreted unless Noam Chomsky's methods and intuitions were adopted (as explicitly witnessed by the appendix of Lenneberg's volume written by Noam Chomsky himself). As for the computational studies, once more, it took Chomsky's formal research to subtract human languages from the alleged supremacy of statistics as suitable models by demonstrating that Markovian chains could not capture the recursive architecture of syntactic dependencies (see the contrast between Shannon 1948 and Chomsky 1956/1965 and the critical opinion in Chomsky - Moro 2023). In fact, Chomsky's first works inaugurated two different fields of research: the neurobiological studies of human language on the one hand and the abstract comprehensive taxonomy of formal grammars and the equivalent class of automata generating them which is traditionally referred to as "Chomsky's hierarchy" (Hopcroft et al. 2006)[3].

Nowadays, both domains of research, the neurobiological and the computational ones, explicitly compete toward exclusive and comprehensive explanatory models for human language: on the one hand, neurobiology, essentially by exploiting neuroimaging, electrophysiology (and, very marginally, genetic data, see Moro 2008/2015), aims at offering reasons as to why only humans developed language and unveil the neural correlates of human language behaviour; on the other hand, within the computational domain, so-called "Artificial Intelligence" as expressed by vLLM, exploiting much more advanced models than Markovian-chains has revived the plausibility of statistical and algorithmic models for human language (see the foundational work in Vaswani et al. 2017). In these new models, sometimes referred to as "Transformers" the crucial failure of previous attempts, essentially constrained on sequential computation has been remedied by adopting a model architecture "eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization." (Vaswani et al. 2017: 2; for illuminating reflections see Bever et al. 2023 and references cited there; for a concrete example of the (still) unsurmountable difference between the structural analyses generated by LLM parsers and syntactic theory see Lorusso et al. 2019).

There's a first fundamental aspect of this tension which should be highlighted: the direction of the two domains of research leads to two non-compatible positions, strictly depending on their different goals. In fact, for example, relying on computational models for clinical studies or therapies would immediately and intuitively sound just paradoxical without the need of further explanations (Moro et al. 2023, Bolhuis 2023).

2. The position of linguistics in the battlefield and of syntax in particular.

Where is linguistics in this scenario? Has it been dissolved into two unreconcilable perspectives? Will linguists need to decide to embrace either domain of research as characterized by different methods and goals and dismiss theoretical linguistics proper? Before addressing some specific issues concerning the linguistic agenda should undertake, I would like to remark on what seems to be an obvious fact for the community of linguists. This fact is so simple that it has hardly ever been explicitly addressed in con–temporary linguistic debates, but on the contrary it is one such that it has never been seriously addressed by neurobiological and computational studies in linguistics. I am referring to the set of robust pervasive systematic boundaries limiting the class of all and only human languages as originally observed by generative grammar (Rizzi 2009) and, *mutatis mutandis*, supported by converging typological studies inaugurated by Joseph Greenberg (see Greenberg 1963 and the illuminating observations in Graffi 1980). In other words, neither domain competing toward a model of human language, i.e. neurobiology and computational studies, have reached descriptive adequacy as stated in Chomsky (1964) and further developed in Chomsky (1965).

Much as in medieval times, these disciplines treat human languages as if there was only one grammar as frequently observed by reporting the famous quotation by Francis Bacon in the equally famous translation by Roman Jakobson: “Grammar is, in its essence, one and the same in all languages, even though it differs in superficial features.” (see Moro 2012/2024 for bibliographical references). This is a very delicate issue and one that must be handled with care: indeed, contemporary linguistics did lead to this very hypothesis as for the existence of a substantially unique human grammar but this conclusion could be reached only throughout a tantalizing theoretical development which led to the “Principles and Parameters framework” as implemented in many subsequent models up to the so-called “Minimalist Program” (Graffi 2001, Demi - Helong 2024). The results of this approach are spectacular: the extension of the empirical domain within and across languages has increased the patrimony of syntactic studies in an unprecedented way, witness monumental results such as those building the Everaert - Van Riemsdijk (2017) collection. They concretely support Giorgio Graffi’s intuition about the present epoch of linguistic studies: “ If, with a little exaggeration, the 19th century has been called "the century of historical-comparative grammar" and the first half of the 20th century, "the age of phonology", one could, with no greater exaggeration, call the second half of this same century "the age of syntax", or "the age of syntactic theories" (Graffi 2001: 309). Synthesizing, Bacon’s view was determined not by a biological or formal argument but rather by a theological one; needless to say, the fact that contemporary neurobiological and computational studies assume that there is one language is rather an oversimplification than a genuine hypo–thesis or a methodological idealization, let alone the result of empirical and theoretical research.

All in all, within these two competing models the systematic differences charac–terizing human languages cannot be the object of empirical research. In fact, so far, only one major syntactic property could be experimentally tested by both systems, namely the hierarchical vs. the linear nature of syntactic dependences showing that the latter does not activate the normal language networks (Moro 2016): surely not a parametric difference since hierarchical structure is obviously common to all and only human languages, as a result of recursion (see Lobina 2017). This is an important property of syntax which has been recognised ever since the first studies in the field but obviously not the only one. In fact, the methodological restrictions imposing any neurobiological paradigm of research to proceed through the comparisons of different tasks or minimally different states makes further steps almost unsurmountable with neuroimaging techniques when it comes to more complex formal syntactic properties, leaving electrophysiological paradigms as the only hopefully suitable ones (for neuro–imaging, see the foundational observations in Friston 1997 and an updated critical vision in Cappa 2012; for electrophysiological methodologies, see Artoni et al. 2020, Cometa et al. 2023, Cometa et al. 2024, Moro 2025, 2016 and references cited there).

There is a last crucial remark which needs to be highlighted: the neurobiological experiments concerning impossible languages mentioned here demonstrated that impossible languages did not activate the natural networks selectively reserved to syntactic computation. These results have been sometimes erroneously interpreted implying that impossible languages could not be learned by humans or that they are more difficult to learn or that they are based on the answer the machine produce to questions (see the different positions in Yang et al. 2025, Ziv et al. 2025, Xiong -Wu 2025, Lan et al. 2024, Chesi 2024 among others): this is plainly false, witness the behavioural measures showing that the number of errors made by the learners was the same in both types of languages. Similarly, these results do not necessarily imply that machines compute possible vs. impossible languages in different ways: what they rather imply is that there isn't any "natural" circuit for possible languages in machines. In fact, if there were one, it would constitute a major problem since it would be very hard to compare the reasons which yield the machines' selective capacity with the evolutionary pressure which led humans to have a uniform hierarchical syntax as opposed to a flat one (see Everaert et al. 2017). Moreover, it seems to me that any consideration about impossible languages in machines based on behavioural data confirms the suspect that the confusion generated in the evaluation of these experiments (and many others) depends on the lack of comprehension of the fundamental distinction between competence and performance, which obviously cannot be immediately transferred to machines.

Summarizing, the brain experiments on possible vs. impossible languages show that the natural networks for language are selectively activated for possible, i.e. hierarchical languages. From a behavioural point of view, instead, the number of errors made while learning impossible languages by humans is the same as for possible ones.

3. An emerging agenda for syntax.

This complex battlefield implicitly and explicitly yields an emergent research program where linguistics maintains its *proprium,* namely its irreducible set of primitive concepts (entities and operations) and defining goals, while not losing the necessity of being a fundamental factor for research in both separate domains of neurobiology and computational theories. It must be noticed that when it comes to neurobiology, the problem of “granularity” as expressed by the influential position of Poeppel (1996) seems in fact to lead to an insufficient, if not misleading therapy, for the question is not quite the “dimension” of the primitive elements (and the associated operations) distinguishing the formal models of language vs. the neurobiological ones. The distance between these models and linguistics is much more profound and could be captured, once more, by adhering to Chomsky’s (1988) analysis which could be dubbed “the Newtonian style of research”: to proceed in science, we can put aside the quest for the explanation of the causes and restrict our goal to require that the theory allows us to draw correlations, simplifications and predictions. The analogy is sharp: as Newton refused to trace gravity back to ether vortices, following the orthodox cartesian view, we should not trace linguistic structures back to what we assume about neurobiological mechanisms underlying neuronal activity (essentially following the spirit of Changeux 1983/1985 and the extreme view in Penrose 1989).

At the same time, we are aware that the linguistic agenda cannot be established only as an answer to external forces and I would like to address my personal view about the fundamental issues which should be explored, both in the methodological and empirical domains of inquiry. Needless to say, linguistics has not yet arrived at the formal maturation mathematics reached in the early XX century and perhaps it never will, but even if there cannot be a “Hilbert list” of modern linguistic problems it seems to me that there are indeed some core questions which should be approached by the scientific community in the next years implicitly manifesting the *proprium* of syntax. Here is a first tentative, personal and certainly not exhaustive list of problems each provided with minimal selective references:

1) provide restrictive theories of functional elements paralleling the development of generative grammar as described in Roberts (1988) and adhering to Chomsky (2024) perspective concerning the quality of expla–nation in (linguistic) theories (see also Demi - Halong 2024 for technical and historical data).

2) investigate the process of externalization including the data proving the existence of acoustic representation in non-acoustic areas during inner speech (Magrassi et al. 2015).

3) characterize the compositional nature of morphology and its relationship with syntax and phonology (Kayne 2019 contra Halle - Marantz 1993).

4) understand and characterise External Merge as expressed in human languages vs. animal language (see Schlenker et al. 2016 and Rizzi 2016) and argumenthood as expressed by Hale-Keyser’s (2002) groundbreaking approach to UTAH.

5) understand the trigger(s) of movement and their relationship with linearization (Moro 1997, 2004, Cinque 2025) including the symmetry-breaking approaches as defined in Dynamic Antisymmetry (Moro 2000,

2009) or, more radically, the very necessity of Internal Merge in all and only human languages (Moro - Roberts 2023).

6) derive typological taxonomies by reformulating the notion of parameter, by avoiding directionality restrictions as proposed in Kayne (2013) in favour of harmony-based treatments (Cinque 2013, Biberauer - Sheenan 2013, Biberauer et al. 2014)

7) extend the domain of empirical recognition across and within languages as assumed by the cartographic approaches pioneered by Cinque - Rizzi (2009) especially in combination with the problem in (1).

8) reformulate and unify the absolute and relative locality conditions, as dubbed by Zeijlstra (2022), expressed in the pre-Minimalist models inclu–ding the key-notions of Superiority, Subjacency and Connectedness (see Chomsky 1973, Kayne 1984, Manzini (1992) and remarks in Moro - Greco - Mocci forthcoming).

9) evaluate parametric models for diachronic dimensions as pioneered in Longobardi - Roberts (2010), Longobardi et al. (2013) and Roberts (2012) ultimately aiming at providing explanatory accounts of language change, which has never really been done (Roberts, p.c.).

10) explore the necessity of maintaining the separation between syntax, morphology and semantics, crucially including the problem of the formal correlates of predication (Moro 2019).

11) establish the fine mapping between pragmatics and syntactic struct–ures across languages (Chierchia 2013, Domaneschi - Bambini 2020).

12) generalize the notion of impossible primitives and impossible combi–natorial operations along the lines discussed in Moro (2016).

We cannot be sure, of course, that any of these issues could be solved or will remain confined in the murky realm of *Ignorabimus!* - to use Emile du Bois-Reymond's famous expression (see Bois-Reymond 1874 and Moro 2024 for a reflection on it) - but what–ever the results will be we can at least be reasonably sure that the relevant questions raised by them could be "genuinely explained" in the sense of Chomsky (2024, 2024b) only if the *proprium* of syntax will be maintained.

More explicitly, dissolving linguistics into other disciplines won't just take us further away from the Newtonian style of research: it would rather quite dramatically avert us from the identification of the class of impossible human languages, a notion which neither neurobiology nor computational theories have (so far) proved to be able to be sensitive to, let alone to capture from a descriptive and explanatory level, and which is the implicit goal of the future agenda. All in all, given the incompatibility of neurobiological models and computational theories when it comes to human language, the acknowledgment of the *proprium* of linguistics would be the only possibility to get out of this battle with a very much hoped threefold victory.

References

1. Artoni, F. - d'Orio, P.- Catricalà, E. - Conca, F. - Bottoni, F. - Pelliccia, V. - Sartori, I. - Lo Russo, G. - Cappa, S. - Micera, S. - Moro, A. (2020) "High gamma response tracks different syntactic structures in homophonous phrases", *Nature Sci Rep* 10**,** 7537.
2. Bever, T. - Chomsky, N. - Fong. S. - Piattelli-Palmarini, M. (2023) "Even deeper problems with neural networks models of language" published on line as a commentary to Bowers, JS et al. (2023) "Deep problems with neural network models of human vision", *Behavioral and Brain Sciences* 46, e385: 1–77. doi:10.1017/S0140525X23001619, e387.
3. Biberauer, Th. – Sheehan, S. (2013) (eds) "Introduction" in *Theoretical approaches to disharmonic word order* (eds) Biberauer, Th. - Sheehan, M. 1–46, Oxford University Press, Oxford, UK.
4. Biberauer, Th. - Holmberg, A. – Roberts, I. (2014) "A syntactic universal and its consequences", *Linguistic Inquiry* ,45: 169–225.
5. Bolhuis, J. - Crain, S. - Fong, S. - Moro, A. (2024) "Three reasons why AI doesn't model human language", *Nature* 627, 489.
6. Bois-Reymond, E., du (1874) "The limits of our knowledge of nature", *Popular Sience Monthly*, pp. 17-32, 369.
7. Cappa, S. F. - Moro, A. - Perani, D. - Piattelli-Palmarini, M. (2000), "Broca's Aphasia, Broca's Area and Syntax: A Complex Relationship," *Behavioral and Brain Sciences* 23: 27–28.
8. Cappa, S.F. (2012) "Imaging semantics and syntax", *NeuroImage*, 2012 Jun;61(2):427-31, doi:10.1016/j.neuroimage.2011.10.006.
9. Changeux, J.-P. (1983/1985), *L'homme neuronal*, Fayard, Paris; *Neuronal Man: The Biology of Mind*, Pantheon Books, New York.
10. Chesi, C. (2024) "Is it the end of (generative) linguistics as we know it?" *Italian Journal of Linguistics*, *36*(2), 3–40.
11. Chierchia, G. (2013) *Logic in grammar*, Oxford University Press, Oxford, England.
12. Cinque, G. - Rizzi, L. (2009) "The Cartography of Syntactic Structures", in Heine, B. - Narrog, H. (eds.) *Oxford Handbook of Linguistic Analysis*, Oxford University Press, pp. 51-65.
13. Cinque, G. (2013) "Deriving word order types: A change of perspective" in *Theoretical approaches to disharmonic word order* (eds) Biberauer, Th. - Sheehan, M. 47-76, Oxford University Press, Oxford, UK.
14. Chomsky, N. (1956/1965), "Three Models for the *Description* of Grammar," *I.R.E. Transactions on Information Theory* IT-2: 113–124; also in R. D. Luce, R. R. Bush, and E. Galanter, eds., *Readings in Mathematical Psychology*, vol. 2, Wiley, New York.
15. Chomsky, N. (1964) *Current issues in linguistic theory*, Mouton, The Hague.

16. Chomsky, N. (1965) *Aspects of the Theory of Syntax*, MIT Press, Cambridge, Massachusetts.
17. Chomsky, N. (1966). *Cartesian linguistics: a chapter in the history of rationalist thought*, Harper & Row, New York; third edition, 2009 Cambridge University Press, Cambridge, UK.
18. Chomsky, N. (1973) "Conditions on transformation", in Anderson, S.R. - Kiparsky, P. *A festschrift for Morris Halle*, Holt, Rinehart and Winston, New York.
19. Chomsky, N. (1988) *Language and Problems of Knowledge: the Managua Lectures*, MIT Press, Cambridge, Mass.
20. Chomsky, N. (2024) "Genuine Explanations", in Bocci, G. - Botteri, D. - Manetti, C. - Moscati, V. (eds.) *Rich Descriptions and Simple Explanations in Morphosyntax and Language Acquisition. In honor of Luigi Rizzi, Oxford University Press. Oxford, UK.*
21. Chomsky, N. (2024)b "The Miracle Creed and SMT", in Greco, M. - Mocci, D. (eds.), *A Cartesian dream: A geometrical account of syntax. In honor of Andrea Moro*, Rivista di Grammatica Generativa/Research Monographs, in Generative Grammar, Lingbuzz Press, Brno, Czech Republic. ISSN 2531-5935.
22. Chomsky, N. – Moro, A. (2023) *The secrets of words*, MIT Press, Cambridge, Mass.
23. Cinque, G. (2025) Five types of Internal Merge and the locus of linearization, ms., Ca' Foscari University, Venice.
24. Cometa, A. - D'Orio, P. - Revay, M. - Bottoni, F. - Repetto, C. - Lo Russo, G. - Cappa - Moro, A. - Micera, S. - Fartoni, F. (2023) "Event-related causality in stereo-EEG discriminates syntactic processing of noun phrases and verb phrases", *J. Neural Eng.*, 20 (2023) 026042.
25. Cometa, A. - Battaglini, C. - Artoni, F. - Greco, M. - Frank, R. - Repetto, C. - Bottoni, F. - Cappa, S. - Micera, S. - Ricciardi, E. - Moro, A. 2024) "Brain and grammar: revealing electrophysiological basic structures with competing statistical models", *Cerebral Cortex* 1, 34 (8).
26. Demi, M. - Helong, T. (2024) *The development of Chomskyan generative grammar: from Standard Theory to the Minimalist Program*, Taylor & Francis, Routledge, New York.
27. Domaneschi, F. - Bambini, V. (2020) "Pragmatic competence", *The Routledge handbook of philosophy of skill and expertise*, Routledge, New York.
28. Everaert, M. - Huybregts, M. - Berwick, R. - Chomsky, N. - Tattersall, I. - Moro, A. - Boluhis, J. J. (2017) "What is language and how could it have evolved?", *Trends in Cognitive Sciences*, 21, 8, p. 569-571.
29. Friederici, A. - Chomsky, N. - Berwick, R. - Moro, A. - Bolhuis, J. (2017) "Language, mind and Brain" *Nature Human Behaviour*, Vol. 1 pp. 713 - 722 DOI: 10.1038/s41562-017-0184-4.
30. Friston, K.J. (1997) "Imaging cognitive anatomy", *Trends in Cognitive Sciences,* vol 1, n°1, April.
31. Graffi, G. (1980), "Universali di Greenberg e grammatica generativa," *Lingua e Stile*, 15: 371–387.
32. Graffi, G. (2001) *200 Years of Syntax. A critical survey*, John Benjamins, Amsterdam, The Netherlands.
33. Greco, M. - Cometa, A. - Artoni, F. - Robert, F. - Moro, A. (2023) "False perspectives on human language: Why statistics needs linguistics", *Frontiers in Language Sciences*, Front. Lang. Sci. 2:1178932.

34. Greenberg, J. H. (ed.) (1963) *Universals of Language*, The MIT Press, Cambridge, MA.
35. Hale, K. - Keyser, J. (2002) *Prolegomenon to a theory of argument structure*, Linguistic Inquiry Monograph, 39, MIT Press, Cambridge, MA.
36. Hawkins, John A. (1983) *Word order universals*, Academic Press, New York, NY.
37. Hopcroft, J. E. - Motwani, R. - Ullmann, J. D. (2006)[3] *Introduction to automata theory, languages, and computation*, Addison-Wesley, Altanta, GA.
38. Kayne, R. (1984) *Connectedness and Binary Branching*, Foris, Dordrecht.
39. Kayne, R. (2013) 'Why Are There No Directionality Parameters?', in Biberauer, Th. – Sheehan, M. (eds), *Theoretical Approaches to Disharmonic Word Order*, Oxford University Press, Oxford, UK.
40. Kayne, R. (2019) "What Is Suppletion? On **Goed* and on *Went* in Modern English", Vol. 117, Issue 3, Special issue: the diachrony of suppletion, pages 434-454.
41. Lan, N. - Chemla, E. - Katzir, R. (2024) "Large Language Models and the argument from the poverty of the stimulus", *Linguistic Inquiry*, doi.org/10.1162/ling_a_00533.
42. Lenneberg, E. (1967) *Biological Foundations of Language*, Wiley, New York.
43. Lobina, D. (2017) *Recursion, A Computational Investigation into the Representation and Processing of Language*, Oxford Studies in Biolinguistics, Oxford University Press, Oxford, England.
44. Longobardi, G. - Roberts, I. (2010) "Universals, diversity and change in the science of language: Reaction to "The Myth of Language Universals and Cognitive ScienceVolume", *Lingua*, Vol. 120, Issue: 12, pp. 2699-2703.
45. Longobardi, G. - Guardiano, C. - Silvestri, G. - Boattini, A. - Ceolin, A. (2013) "Toward a syntactic phylogeny of modern Indo-European languages", *Journal Journal of Historical Linguistics* 3:1 (2013), 122–152.
46. Lorusso, P. - Greco, M. - Chesi, C. - Moro, A. (2019). *Asymmetries in Extraction From Nominal Copular Sentences: a Challenging Case Study for NLP Tools*. In *Proceedings of the Sixth Italian Conference on Computational Linguistics* (CLiC-it 2019). In CEUR workshop Proceedings - ISSN:1613-0073.
47. Magrassi, L. - Aromataris, G. - Cabrini, A. - Annovazzi-Lodi, V. - Moro, A. (2015) "Sound Representation in Higher Language Areas during Language Generation," *Proceedings of the National Academy of Sciences* 112(6): 1868–1873.
48. Manzini, R. (1992), *Locality*, MIT Press, Cambridge, MA.
49. Moro, A. (1997) "Dynamic Antisymmetry: movement as a symmetry-braking phenomenon", *Studia Linguistica*, 51, pagg. 50-76.
50. Moro, A. (2000) *Dynamic Antisymmetry*, MIT Press, Cambridge, MA.
51. Moro, A. (2004) "Linear compression as a trigger for movement, in Riemsdijk, H. van - Breitbarth, A (eds.) *Triggers*, Mouton de Gruyter, Berlin.
52. Moro, A. (2008/2015) *The boundaries of Babel,* II edition. The MIT Press, Cambridge MA, USA.
53. Moro, A. (2009) *Rethinking Symmetry: A Note on Labelling and the EPP*, in Cotticelli Kurras, P – Tomaselli, A. (eds.), *La grammatica tra storia e teoria. Scritti in onore di Giorgio Graffi*, Edizioni dell'Orso, Alessandria, pp. 129-31; reprinted in Moro, A. (2013) *The Equilibrium of Human Syntax. Symmetries in the brain*, Leading Linguists Series, Routledge, New York.
54. Moro, A. (2016) *Impossible Languages*, MIT Press, Cambridge, Mass.
55. Moro, A. (2019) "The geometry of predication", *Phil. Trans. R. Soc.* B 375.

56. Moro, A. (2012/2024) *Parlo dunque sono. 17 Istantanee sul linguaggio*, II edizione, Adelphi, Milano; trad. ing. 2016 *I Speak, therefore I am*, Columbia University Press, New Yorks.
57. Moro, A. (2025) “The sound of silence: on inner speech and the architecture of the language faculty”, *Lingue e Linguaggio*, XXIV.1, 11–19.
58. Moro A. et al. (2001) Syntax and the brain: disentangling grammar by selective anomalies, *Neuroimage* 13(1): 110-118.
59. Moro, A. - Greco, M - Cappa, SF (2023) “Large languages, impossible languages and human brains”, *Cortex*, Oct; 167:82-85. doi: 10.1016/j.cortex.2023.07.003. Epub 2023 Jul 17. PMID: 37540953.
60. Moro, A. - Greco, M. - Mocci. D. (forthcoming) “Revisiting locality: two case studies from Italian, *Isogloss*.
61. Moro, A. - Roberts, I. (2023) “The duality of syntax: unlabelled structures, labelling and linearization”, *NLLT, https://doi.org/10.1007/s11049-023-09588-z*
62. Murphy, E. (2020) *The oscillatory nature of language*, Cambridge University Press, UK.
63. Musso, M. - Moro, A. - Glauche. V. - Rijntjes, M. -Reichenbach, J. - Büchel, C. - Weiller, C. (2003) “Broca’s area and the language instinct,” *Nature neuroscience*. vol.6, pp. 774-781.
64. Penrose, R. (1989) *The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics*, Oxford Univ. Press, Oxford, England.
65. Poeppel, D. (1996) “Neurobiology and linguistics are not yet unifiable”, *Behavioral and Brain Sciences*, 19(4), 642-643.
66. Rizzi, L. (2009) “The discovery of language invariance and variation, and its relevance for the cognitive sciences”, *Behavioral and Brain Sciences*, 32:467-468 Cambridge University Press.
67. Rizzi, L. (2016) “Monkey morpho-syntax and merge-based systems” ,Theoretical Linguistics 2016; 42(1-2): 139–145.
68. Roberts, I. (1988), “From Rules to Constraints,” *Lingua e Stile* [Language and style] 23: 445–464.
69. Roberts, Ian (2012) “Macroparameters and minimalism: A programme for comparative research”, in *Parameter theory and linguistic change* (eds.) Galves, C. - Cyrino, S. - Lopes, R. - Sandalo, F. - Avelar, J., pp. 320–334. Oxford University Press, Oxford, UK.
70. Shannon, C. E. (1948) “A mathematical theory of communication”, *Bell System Technical Journal,* vol. 27, pp. 379-423 and 623-656, July and October.
71. Schlenker, P. - Chemla, E. - Schel, A. M. – Fuller, J. – Gautier, J.-P. – Kuhn, J. – Vaselinovič, D. – Arnold, K. – Cäsar, C. – Keenan, S. – Lemasson, A. Ouattara, K. – Ryder, R. – Zuberbühler, K. (2016) “Formal monkey linguistics”, *Theoretical Linguistics* 2016; 42(1-2): 1–90.
72. Tettamanti, M. - Alkadhi, H. - Moro, A. - Perani, D. - Kollias, S. - Weniger D. (2002) “Neural correlates for the acquisition of natural language syntax”, *NeuroImage*, 17, 700-709.
73. Vaswani, A. – Shazer, N. - Parmar, N. - Uszkoreit, J. - Jones, L. -Gomez, A. - Kaiser, Ł. - Polosukhin, I. (2017) “Attention is all you need”, in Guyon, I. - Von Luxburg, U. – Bengio, S. – Wallach, H. - Fergus, R. – Vishwanathan, S. – Garnett, R. (ed.) 31st Conference on Neural Information Processin Systems (NIPS)

Advances in Neural Information Processing Systems, Vol. 30. Curran Associates, Inc. arXiv:1706.03762.

74. Yang, X. – Aoyama, T. – Yao, Y. – Wilcox, E. (2025) "Anything goes? A crosslinguistic study of (Im)possible language Learning in LMs, arXiv:2502.18795v2.
75. Xiong, Y. – Wu, Shiyu, W. (2025) "Do large language models learn like humans: Interleaved and spaced practice in morphological learning", *Acta Psychologica*, Volume 260, October 2025, 105518.
76. Zeijlstra, H. (2020) "Absolute vs. relative locality", 46th Generative Grammar in the South Conference, 19-21 May, University of Stuttgart.
77. Ziv, I. – Chemmla, E. – Lan, N. – Katzir, R. (2025) "Large Language Models as Proxies for Theories of Human Linguistic Cognition", arxiv.org/html/2502.07687v1.